Social Media Intelligence (SOCMINT) Aggregation

Deriving Pattern of Life from Fragmented Digital Traces

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Abstract

This project investigates the application of Open-Source Intelligence (OSINT) to synthesize fragmented digital footprints into cohesive Pattern of Life (POL) profiles, a phenomenon known as the "Mosaic Effect." A custom, offline analytical dashboard was developed using Python to empirically demonstrate this. The system ingests decoupled, scraped social media data and applies programmatic heuristics, including VADER sentiment analysis, spatial categorization, and temporal windowing, to identify behavioral anomalies. To strictly adhere to GDPR regulations and ethical academic standards, the research was conducted entirely offline using synthetic datasets. The results successfully prove that isolated, seemingly benign data points can be programmatically aggregated to reveal sensitive, actionable intelligence about a target's lifestyle, associations, and vulnerabilities.

Introduction

Contemporary online life is characterized by the continuous generation of unintentional data traces, often referred to as "digital exhaust." While a single piece of data may not be threatening, when these traces are multiplied over time, they form a mosaic. An adversary can compile these fragments to recreate a detailed user profile, bypassing privacy controls [4].

This systemic oversharing provides the foundational data required for OSINT PoL profiling [1]. The central question addressed by this study is to determine the extent to which a post-acquisition OSINT analysis engine can form a rich Pattern of Life, place profile, and sentiment profile of an individual solely from publicly available social media traces.

Project Idea & Objectives

The primary goal is to create, implement, and test a Python-based OSINT application that ingests, normalizes, and combines pre-collected datasets to demonstrate a high, but extremely underrated, confidence threat. Key objectives include:

1. Develop a post-acquisition ingestion module capable of securely processing standard offline datasets (e.g., CSV, XLSX) to ensure strict Operational Security (OPSEC).
2. Use timestamp analysis to generate PoL heatmaps.
3. Perform sentiment analysis of text in social media with an adequate lexicon-based model (VADER).
4. Implement a Social Network Analysis (SNA) module utilizing graph theory to map associated entities and highlight secondary data leakage.

Methodology & Architecture

To mitigate the inherent volatility of modern social media platforms and aggressive anti-scraping countermeasures, raw data acquisition is strategically decoupled from the primary analysis tool. Data extraction is delegated to an established third-party service (Apify) that exports standardized offline datasets.

The offline analytical pipeline relies on several core modules:

- **Data Normalization:** Converts all timestamps into a unified date and time format, adjusting them to Coordinated Universal Time (UTC).
- **Temporal Engine:** Calculates daily frequencies, active hours, and infers user time zones to establish a routine baseline.
- **Sentiment Engine:** Evaluates post captions using VADER to assign emotional polarity scores. A lightweight heuristic fallback lexicon is hard-coded to ensure fault tolerance in air-gapped environments.
- **Geospatial & SNA:** Extracts mentions, tags, and spatial entities to generate relational graphs using NetworkX.

Results & Findings

The temporal analysis reveals that even small datasets can portray consistent behavioral patterns when subjected to systematic OSINT aggregation.

Dimension	Analytical Focus	OSINT Data Sources
Spatial	Geographic regularity	EXIF GPS, geotags
Temporal	Time-based routines	Post timestamps
Behavioral	Expressive patterns	Text content, sentiment

Table 1: Pattern of Life Classification Dimensions in OSINT

Sentiment analysis outcomes added an interpretive layer complementary to time dynamics. The sentiment visualizations on a daily average scale were particularly helpful to display momentary emotional variations.

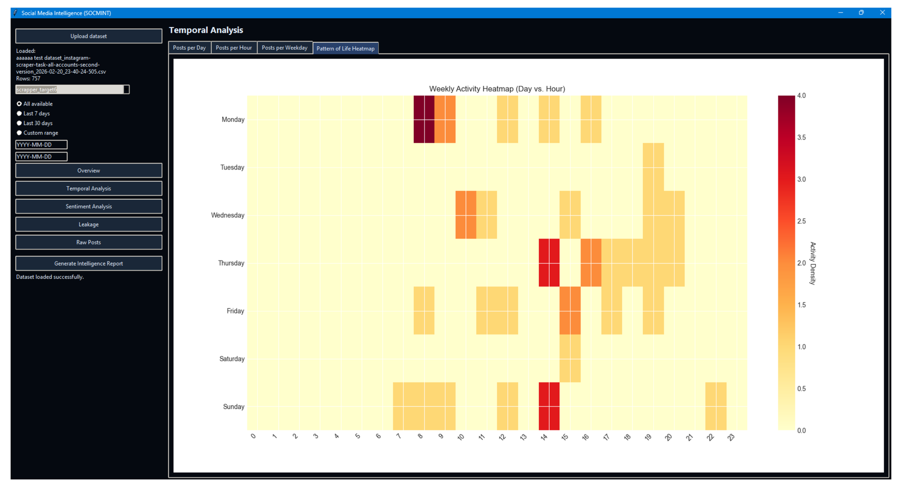

Figure 1: Pattern of Life Activity Heatmap

Regarding spatial data, the image leakage evaluation yielded minimal findings regarding hidden GPS metadata, which accurately reflects modern platform operational security (OPSEC) where major social networks actively strip EXIF data.

Automated Anomaly Detection

Anomaly detection is robustly operationalized through an automated heuristic engine. Rather than relying solely on manual observation, the system programmatically synthesizes deviations:

Anomaly Type	Programmatic Trigger Condition
Travel Anomaly	Sudden shift in inferred physical timezone
Routine Disruption	Post timestamp falls between 00:00 and 06:00
Mood Drop	Post sentiment < -0.5 AND rolling average > 0

Table 2: Automated Anomaly Detection Triggers

Conclusions

- Results show that moderate and even minimal levels of social media use can yield meaningful behavioral patterns through systematic analysis.
- By visualizing incidental leakage and routine predictability via the offline GUI, the tool tangibly highlights the severe privacy risks of continuous digital exhaust, successfully serving as an effective educational red-team artifact.
- To guarantee legal compliance and ethical integrity (UK DPA / GDPR), this project successfully utilized synthetic target accounts, proving technical feasibility without exposing real individuals to privacy risks.

Limitations & Further Work

Relying on synthetic data restricts the generalizability of the results, as real-world social media behavior is richer and less predictable. Furthermore, the system's heuristic Natural Language Processing (NLP) for location categorization relies on static keyword matching, which lacks true semantic understanding and struggles with idioms.

Future iterations should integrate Named Entity Recognition (NER) models (such as spaCy) or Contextual Word Embeddings (like BERT) to semantically distinguish between a physical location and a conversational idiom, drastically reducing false positives [2, 3]. Additionally, predictive machine learning modeling could be applied to forecast future behavior.

References

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